Audio Feature Extraction Summary

Module: Audio Feature Extraction for AI-Based Dementia Detection Using Speech Analysis

This module extracts acoustic features from speech recordings in the Pitt Dementia Corpus. These numerical features are later combined with linguistic features and used to train a machine learning model for dementia detection.

Python Libraries Installed

pip install librosa numpy pandas scipy soundfile

Imported Libraries

```
import os
import librosa
import numpy as np
import pandas as pd
```

Main Code Structure

```
DATASET_PATH = "Dataset/Audio"
for label in ["Control","Dementia"]:
  label_path = os.path.join(DATASET_PATH,label)
  # Read each audio file
  y, sr = librosa.load(file_path, sr=16000)
  mfcc = librosa.feature.mfcc(y=y,sr=sr,n_mfcc=13)
  duration = librosa.get_duration(y=y,sr=sr)
  energy = np.mean(y**2)
```

Features Extracted

• MFCC (13 coefficients)
• Pitch
• Energy
• Speech Duration

Output

The extracted features of every audio file are stored in **features.csv**. This CSV is later merged with linguistic features generated from transcripts for machine learning model training.

Workflow

Dataset → Audio Preprocessing → Feature Extraction → features.csv → Feature Fusion → Model Training → Prediction